Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Danfei Xu	**Email**	danfei@gatech.edu
Institute Name	Georgia Institute of Technology			
Address	266 Ferst Dr. NW, Atlanta, GA 30332			

2. Technology Overview

Title of Proposed Technology	Safe and Faster-than-expert Imitation Learning for General-Purpose Manipulation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	*Robotics/ Automation Machinery*	Robot Vision and Production Automation Technologies
Concept of Proposed Technology	The proposed technology develops a **safe and faster-than-expert imitation learning (IL) stack** for **general-purpose robot manipulation**, targeting deployment on real robot hardware in cluttered, human-shared workspaces (e.g., light manufacturing, logistics, lab automation, and service settings). The core idea is to combine:			

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	1. **Speed-adaptive execution** that enables an offline-learned visuomotor policy to run **faster than the demonstrations**, improving throughput while remaining stable under dynamics shift and real-world latency (SAIL [1] achieves up to **4×** speedup in simulation and **3.2×** on real robots). 2. **Hard-constraint safety enforcement** via a reachability-based safety filter that validates planned motion in continuous time and attaches a failsafe backup plan, achieving **0% collision rates** in evaluated tasks while remaining real-time on hardware (RAIL [2]). Together, this yields a manipulation system that is both **production-relevant (throughput)** and **deployment-relevant (hard safety constraints)**, without requiring online RL fine-tuning. **Key features** • **Faster-than-demonstration execution for imitation policies** through a full-stack speed adaptation framework (consistency-preserving inference, controller-invariant targets, adaptive speed modulation, and latency-aware action scheduling). • **Reachability-based safety filter for hard constraints** that enforces continuous-time collision avoidance (and other constraints) at inference time while staying compatible with state-of-the-art IL policies. • **Composable deployment pipeline** that ports to new robot embodiments and controllers via controller-aware policy targets and a modular safety supervisor layer. • **Real-time operation on hardware** with explicit handling of sensing-to-action delay and continuous-time plan validation.
Rationale for Joint R&D Project	A joint R&D project is essential because the proposed technology is a **software–hardware co-design problem**: • **Speedup depends on hardware control and latency**: achieving faster-than-demo execution requires tight integration with the robot's low-level control stack, timing, sensors, and compute pipeline (SAIL explicitly accounts for controller behavior and system latency). • **Safety depends on embodiment-specific geometry and sensing**: hard constraint enforcement requires accurate forward occupancy / collision models and runtime state estimation to validate trajectories and execute failsafes (RAIL's reachability-based filter is designed to remain real-time on manipulators). • **Industrial translation requires validation under realistic constraints**: SMEs can provide target platforms, integration engineering, and deployment testbeds to verify throughput gains and safety compliance in representative operational environments.
Key Words	1. Faster-than-Demonstration Imitation Learning

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	2. Reachability-Based Safety Filter 3. Hard-Constraint Collision Avoidance 4. Latency-Aware Real-Time Policy Execution 5. General-Purpose Robot Manipulation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this project is to develop and validate a **high-throughput, safety-guaranteed manipulation execution system** that can deploy imitation-learned policies on real robots without sacrificing reliability. Specific objectives include: • Enabling imitation policies to execute **2–3× faster than human demonstrations** while maintaining task success [1]. • Enforcing **hard safety constraints** such as collision avoidance during accelerated execution, with zero tolerance for safety-critical failures [2]. Delivering an integrated prototype that runs in real time on an SME-provided robot platform.			
R&D Target Specifications	**Execution Speed** • ≥2× average speedup over demonstration time across benchmark manipulation tasks • Stretch goal: ≥3× speedup on selected tasks **Task Performance** • ≥80% task success rate under accelerated execution **Safety** • 0 safety-critical collisions during evaluation under enforced constraints **Real-Time Performance** • Stable closed-loop execution without action starvation under realistic sensing-to-actuation latency			
R&D Plan	**Year 1: Algorithm Development and Simulation Validation** • Implement speed-adaptive execution mechanisms for imitation policies, including controller-aware targets and latency-aware scheduling [1]. • Implement reachability-based safety filtering with continuous-time validation and backup planning [2]. • Validate speed–success–safety tradeoffs in simulation benchmarks. **Year 2: Hardware Integration and System Co-Design** • Integrate the full system on an SME manipulation platform. • Calibrate controller models, latency bounds, and safety verification models for the target embodiment. • Conduct extensive real-robot experiments to validate throughput, success rate, and safety compliance. **Year 3: Prototype Refinement and Pre-Commercial Validation** • Expand task coverage and environmental diversity. Perform long-horizon and stress testing under safety-critical scenarios. • Deliver a deployment-ready prototype and technical documentation for technology transfer.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Develop speed-adaptive imitation learning execution algorithms [1]. • Develop reachability-based safety supervision and continuous-time constraint enforcement [2]. • Design evaluation protocols and performance metrics. • Lead algorithmic analysis and system-level validation.
Expected R&R of Korean SME Partner	• Provide robot hardware, sensors, and system integration support. • Implement control interfaces, safety interlocks, and deployment infrastructure. • Execute large-scale experimental validation in representative operational environments. • Lead productization and commercialization planning.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• A deployable manipulation system that combines **high throughput** with **hard safety guarantees**. • Demonstrated real-world performance improvements over standard imitation learning baselines. • A modular software stack ready for integration into industrial robot products. Impact: • Enables Korean SMEs to deploy learning-based manipulation systems with reduced safety risk. • Improves productivity in automation scenarios where speed and safety are both critical. • Lowers the barrier to commercial adoption of imitation learning in real-world robotics.

6. References

[1]SAIL:Faster-than-Demonstration Execution of Imitation Learning Policies, Nadun Ranawaka Arachchige, Zhenyang Chen, Wonsuhk Jung, Woo Chul Shin, Rohan Bansal, Pierre Barroso, Yu Hang He, Yingyang Celine Lin, Benjamin Joffe, Shreyas Kousik, Danfei Xu, Conference on Robot Learning 2025

[2] RAIL: Reachability-Aided Imitation Learning for Safe Policy Execution, Wonsuhk Jung, Dennis Anthony, Utkarsh A. Mishra, Nadun Ranawaka Arachchige, Matthew Bronars, Danfei Xu, Shreyas Kousik, Conference on Robot Learning 2024

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☐ **Terms:**

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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